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Where Minds Pollinate

Class 9
Chemistry
Chapter 1
NOTES

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Fundamentals of Chemistry

Activity: Q1: "Branches of chemistry"

→ **Chemistry:** chemistry is defined as the science that examines the materials of the universe & changes that these materials undergo.

→ The study of chemistry is commonly divided into eight major branches.

1- **Physical chemistry:**

The branch of chemistry that deals with laws & theories to understand physical chemistry.

2- **Organic chemistry:**

The branch of chemistry that deals with substances containing carbon except carbonates, bicarbonates, oxides & carbides.

3- **Inorganic chemistry:**

The branch of chemistry that deals with elements & their compounds except organic compounds is called Inorganic chemistry.

4- **Biochemistry:**

The branch of chemistry that deals with (elements) physical & chemical changes that occur in living organisms is called Biochemistry.

6- Industrial chemistry: *

The branch of chemistry that deals with the methods & use of technology in the large-scale production of useful substances is called industrial chemistry.

6- Environmental chemistry: *

The branch of chemistry that deals with the chemicals & toxic substances that pollute the environment & their adverse effects on human beings is called environmental chemistry.

7- Analytical chemistry: *

The branch of chemistry that deals with the methods & instruments for determining the composition of matter is called analytical chemistry.

8- (N^e) Nuclear chemistry: *

The branch of chemistry that deals with the changes that occur in atomic-~~nes~~ nuclei is called Nuclear chemistry.

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14th March-24

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Example: 1 * 1 ✓

1- Identify the branch of chemistry in each of the following example.

1- Photosynthesis produce glucose & oxygen from water & carbon dioxide in presense of chlorophyll & Sunlight.

A: Bio chemistry; Since photosynthesis is a chemical reaction that occurs in plants.

2- Plantation help in over coming green house effect.

A: Environmental chemistry; Since green house effect is an enviromental pollution.

3- Habers process converts large quantities of nitrogen & hydrogen in ammonia (NH_3)

A: Industrial chemistry; Since large scale produce any substance is the subject of industrial chemistry.

4- Ammonia is a colour less gas pungent irritating odor which is highly Solable in water.

A: Inorganic chemistry; Since it deals with the properties of inorganic compounds.

5- A chemist performed an experiment to cheak the percentage purity of a sample of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)

A: Analytical Chemi

of compound, when it is organic or inorganic in nature.

An analyst determines that NO_3 is responsible for acid rain.

Environmental chemistry: Since acid rain is an environmental problem.

Some other branches of chemistry are discuss below.

9. Medicinal chemistry: *

The branch of chemistry that deals with the study of drugs of interaction between drugs & biological targets as well as the development of new medicinal agents.

10. Polymer chemistry: *

The branch of chemistry that focus on the study of polymer, their types, ^{properties} ~~properties~~, importance & type of polymerization. is called polymer chemistry. Example of synthetic polymer include nylon bearings, plastic bags, polythene, polyester, epoxy etc.

11. Geo chemistry: *

The branch of chemistry that deals with the study of chemical composition, distribution & transformation of elements & compound in the earth crust such as rock, minerals, soil, water & the atmosphere.

— °° (Question & Answers) °° —

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Q2 What are the daily life application of chemistry?

A1. Organic chemistry: To treat diseases, organic chemistry synthesize new medicines that interact with specific targets, like proteins or enzymes.

a. Inorganic chemistry: Lithium-ion (Li-ion) batteries are used as rechargeable batteries for electronics, toys, wireless headphone, hand hold power tools, small & large appliances, electrical storage devices & electric vehicles.

3. Analytical chemistry: Forensic chemistry is the application of analytical chemistry. It involves the examination of physical traces, such as body fluids, bones, fibre, & drugs. It can be used to identify an unknown compounds. For example drugs are often found, various colored powders & are analysed to determine their context.

4. Physical chemistry: Physical chemistry is a part of our everyday life. The batteries in our vehicles are built on the principle of electrochemistry. ✓

5. Environmental chemistry: Environmental chemistry is used to protect water that has poisoned by soil & dust by using different methods example: sedimentation, filtration, & disinfection.

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—°(Questions & Answers)°—

Q3 Define Science, technology & Engineering.

Ans: Science:-

"Science is the systematic study to explore the natural world. Science intends to recognize the fundamental principles & process of the natural world."

Technology:-

"The integration of scientific knowledge from needs is known as technology." It helps in discovery and development of new materials, in improving quality of products.

Engineering:-

"Engineering is the application of scientific principles to construct and improve systems, structure and machines" chemical engineers develop & design manufacturing process for the production of chemical fuels, foods, medicine, polymer, paper etc.

APPLICATION OF SCIENCE & TECHNOLOGY & ENGINEERING :-

- 1) Investigation rusting of iron:- Scientists investigate to find the chemical reaction that occurs between iron, water & oxygen that cause rust to form. Experiments could be conducted to understand the factors that influence (their) this process & help develop strategies to prevent rust.
- 2) Harvesting Solar Energy:- Scientist may study the principle of photovoltaic cells to understand how sunlight can be converted into electricity. Technologists can develop solar panels. Engineers design & implement large scale renewable energy system. For instance, an electrical Engineer might design the wiring & connections of solar power plant, Civil Engineer may design the infrastructure.
- 3) Designing a water filtration system:- During designing of a water filtration plant, chemical engineers help to design & the process to remove contaminants from water, while mechanical engineers design physical components. Together, they create a solution to clean, drinking water.

iv) Organic chemistry: Scientists study about carbohydrates to learn about the production of oil. Food technologists extract oil from seeds. Chemical engineers design oil production equipment & processes so that oil is produced efficiently, & safely for cooking.

v) Plastic Bags: Scientists study the small building blocks known as monomers. ~~combine~~, they form long chains known as polymers. One of these long chains is polythene, which is one of the many polymers found in plastic bags. Engineers & technicians discover to create bags.

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